

CS 228 : Logic in Computer Science

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Recap

- ▶ We are interested in checking whether a FO formula over words is satisfiable. For this, we plan to convert it an “equivalent” automaton, and check if some good state is reachable in this constructed automaton.
- ▶ FO-definable languages
- ▶ Regular languages

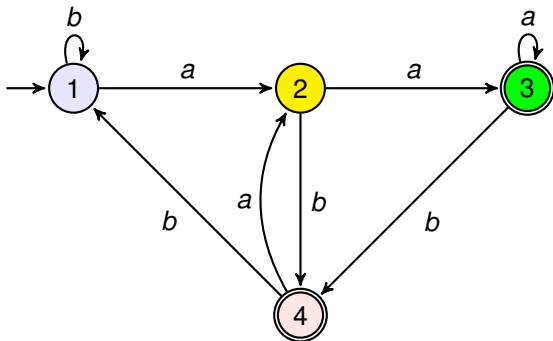
Is it Regular? Is it FO-definable?

$\Sigma = \{a, b\}$. Consider the following languages $L \subseteq \Sigma^*$:

- ▶ Right before the last position is an a :

Examples : $ab, babbaa, bbab$

Non examples : ba, bb, aba



$$\exists x[Q_a(x) \wedge \exists y(S(x, y) \wedge \forall z(z \leq y))]$$

Deterministic Finite Automata

- ▶ Every state on every symbol goes to a unique state
 - ▶ $\delta : Q \times \Sigma \rightarrow Q$ is a transition function
- ▶ Given a string $w \in \Sigma^*$ and a state $q \in Q$, iteratively apply δ
 - ▶ $w = aab$
 - ▶ $\delta(q, a) = q_1$,

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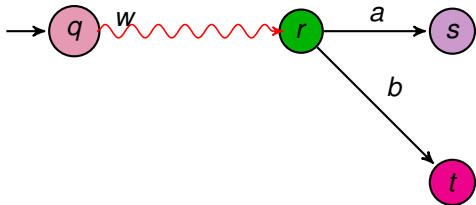
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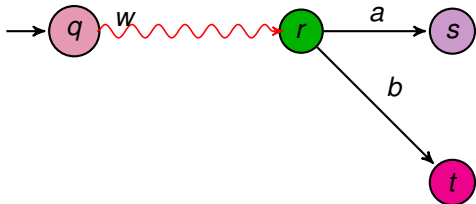
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 $\delta(\delta(\delta(q, a), a), b) = \delta(\delta(q_1, a), b) = \delta(q_2, b) = q_3$
 - ▶ $\hat{\delta} : Q \times \Sigma^* \rightarrow Q$ extension of δ to strings
 - ▶ $\hat{\delta}(q, \epsilon) = q$
 - ▶ $\hat{\delta}(q, wa) = \delta(\hat{\delta}(q, w), a)$

DFA : Transition Function on Words



DFA : Transition Function on Words



- ▶ $\hat{\delta}(q, wa) = s = \delta(\hat{\delta}(q, w), a) = \delta(r, a)$
- ▶ $\hat{\delta}(q, wb) = t = \delta(\hat{\delta}(q, w), b) = \delta(r, b)$

DFA Acceptance

- ▶ $w \in \Sigma^*$ is accepted iff $\hat{\delta}(q_0, w) \in F$
- ▶ $w \in \Sigma^*$ is rejected iff $\hat{\delta}(q_0, w) \notin F$
- ▶ Any string $w \in \Sigma^*$ is either accepted or rejected by a DFA A
- ▶ $L(A) = \{w \in \Sigma^* \mid \hat{\delta}(q_0, w) \in F\}$
- ▶ $\Sigma^* = L(A) \cup \overline{L(A)}$

Partitioning Σ^* into Buckets

Consider $L = a^*b^*$. How does the DFA look?

Partitioning Σ^* into Buckets

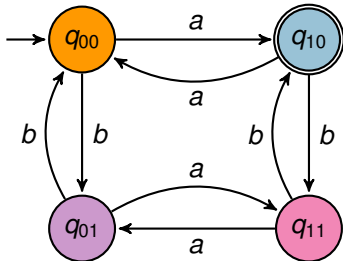
Consider $L = a^*b^*$. How does the DFA look?

- ▶ $[\epsilon] = a^*$
- ▶ $[b] = a^*b^+$
- ▶ $[ba] = a^*b^+a\Sigma^*$

DFA States

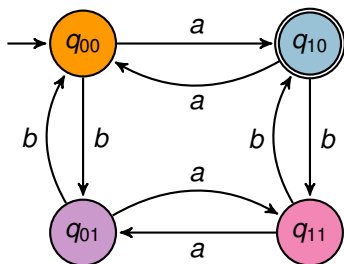
- ▶ Each state is a **bucket** holding infinitely many words
- ▶ Thus we have good and bad buckets
- ▶ The buckets partition Σ^*
- ▶ **Good buckets** determine the language accepted by the DFA
- ▶ Words that land in bad buckets are not accepted by the DFA

Language Acceptance : Proof



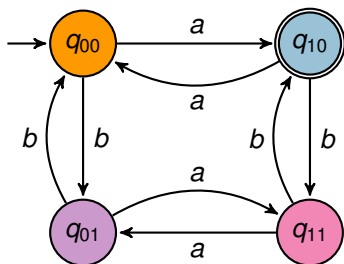
- Prove by induction on $|w|$

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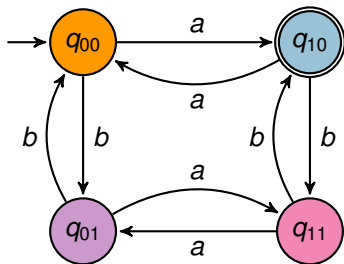
- ▶ Prove by induction on $|w|$
- ▶ Base case : For $|w| = \epsilon$, $\hat{\delta}(q_{00}, \epsilon) = q_{00}$

Language Acceptance : Proof



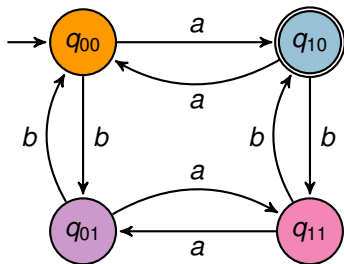
- ▶ Prove by induction on $|w|$
- ▶ Base case : For $|w| = \epsilon$, $\hat{\delta}(q_{00}, \epsilon) = q_{00}$
- ▶ Assume the claim for $x \in \Sigma^*$, and show it for xc , $c \in \{a, b\}$.

Language Acceptance : Proof



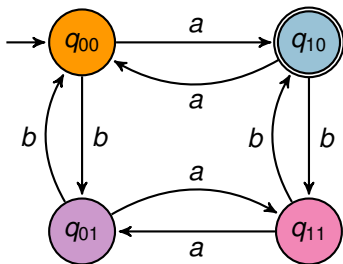
► $\hat{\delta}(q_{00}, xc) = \delta(\hat{\delta}(q_{00}, x), c)$

Language Acceptance : Proof



- ▶ $\hat{\delta}(q_{00}, xc) = \delta(\hat{\delta}(q_{00}, x), c)$
- ▶ By induction hypothesis, $\hat{\delta}(q_{00}, x) = q_{ij}$ iff
 - ▶ parity of i and $|x|_a$ are the same
 - ▶ parity of j and $|x|_b$ are the same

Language Acceptance : Proof



- ▶ Case Analysis : If $|x|_a$ odd and $|x|_b$ even, then $i = 1, j = 0$
 - ▶ $\delta(q_{10}, a) = q_{00}, \delta(q_{10}, b) = q_{11}$
 - ▶ $|xa|_a$ is even and $|xa|_b$ is even
 - ▶ $|xb|_a$ is odd and $|xb|_b$ is odd
- ▶ Other Cases : Similar
- ▶ $\hat{\delta}(q_{00}, x) = q_{10}$ iff $|x|_a$ odd and $|x|_b$ even

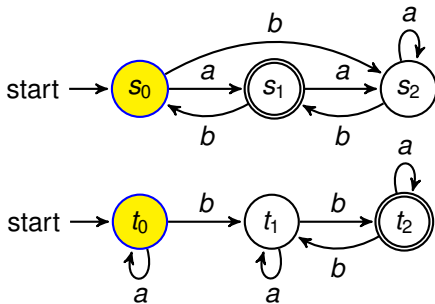
Closure Properties : DFA

Closure under Complementation

- ▶ If L is regular, so is $\bar{L}$
 - ▶ Let $A = (Q, q_0, \Sigma, \delta, F)$ be the DFA such that $L = L(A)$
 - ▶ For every $w \in L$, $\hat{\delta}(q_0, w) = f$ for some $f \in F$
 - ▶ For every $w \notin L$, $\hat{\delta}(q_0, w) = q$ for some $q \notin F$
 - ▶ Construct $\bar{A} = (Q, q_0, \Sigma, \delta, Q - F)$
 - ▶ $w \in L(\bar{A})$ iff $\hat{\delta}(q_0, w) \in Q - F$ iff $w \notin L(A)$
 - ▶ $L(\bar{A}) = \bar{L(A)}$

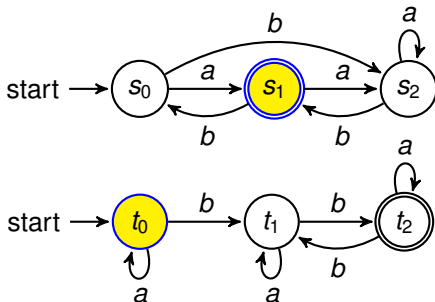
Closure under Intersection

► *aaab*



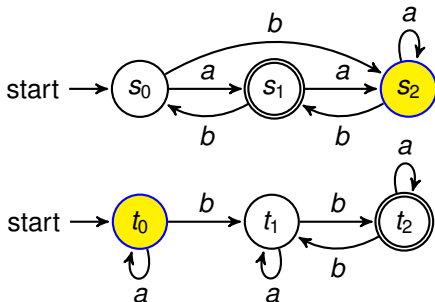
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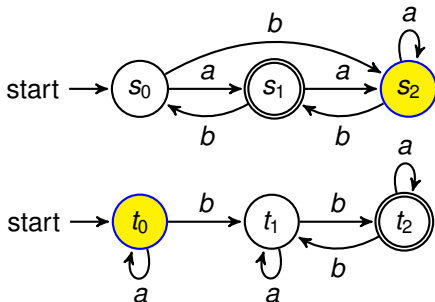
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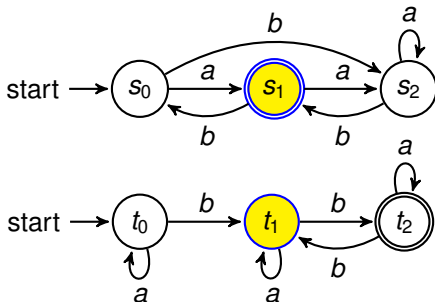
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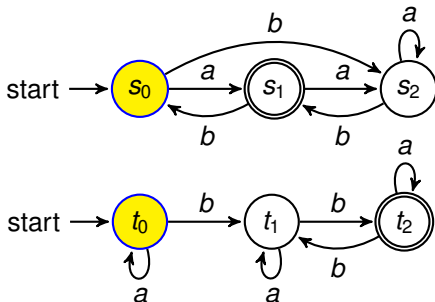
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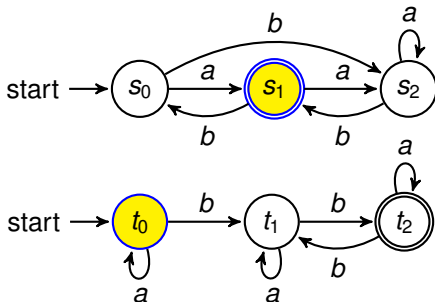
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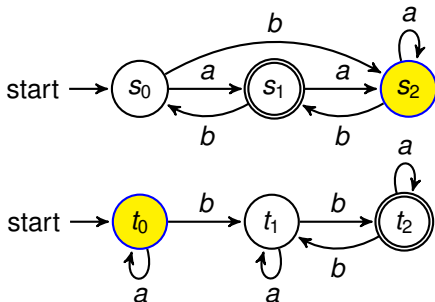
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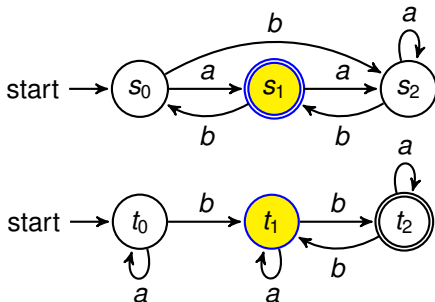
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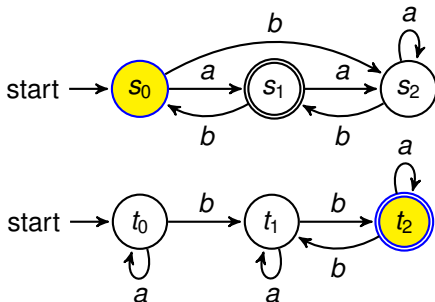
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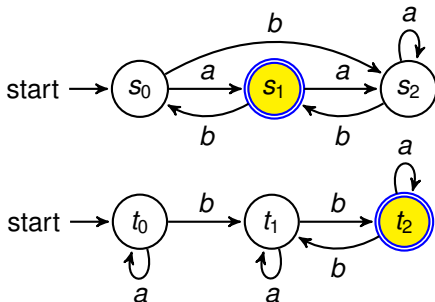
Closure under Intersection

► *aabba*



Closure under Intersection

► *aabb***a**



Closure under Intersection

- ▶ $A_1 = (Q_1, \Sigma, \delta_1, q_0, F_1)$
- ▶ $A_2 = (Q_2, \Sigma, \delta_2, s_0, F_2)$
- ▶ $A = (Q_1 \times Q_2, \Sigma, \delta, (q_0, s_0), F),$
 - ▶ $\delta((q, s), a) = (\delta_1(q, a), \delta_2(s, a))$
 - ▶ $F = F_1 \times F_2$

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 - ▶ $\delta((q, s), a) = (\delta_1(q, a), \delta_2(s, a))$
 - ▶ $F = F_1 \times F_2$
- ▶ Show that for all $x \in \Sigma^*$, $\hat{\delta}((p, q), x) = (\hat{\delta}_1(p, x), \hat{\delta}_2(q, x))$

$x \in L(A)$ iff $\hat{\delta}((q_0, s_0), x) \in F$

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 $\hat{\delta}_1(q_0, x) \in F_1$ and $\hat{\delta}_2(s_0, x) \in F_2$ iff $x \in L(A_1)$ and $x \in L(A_2)$

Closure under Union

- ▶ $A_1 = (Q_1, \Sigma, \delta_1, q_0, F_1)$
- ▶ $A_2 = (Q_2, \Sigma, \delta_2, s_0, F_2)$
- ▶ $A = (Q_1 \times Q_2, \Sigma, \delta, (q_0, s_0), F),$
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 - ▶ $\delta((q, s), a) = (\delta_1(q, a), \delta_2(s, a))$
 - ▶ $F = (F_1 \times Q_2) \cup (Q_1 \times F_2)$
- ▶ Show that for all $x \in \Sigma^*$, $\hat{\delta}((p, q), x) = (\hat{\delta}_1(p, x), \hat{\delta}_2(q, x))$

$x \in L(A)$ iff $x \in L(A_1)$ or $x \in L(A_2)$